GROWTH HORMONE & ANABOLIC SIGNALING

Ipamorelin — the full review

SELECTIVE GROWTH HORMONE SECRETAGOGUE

This is the deeper look behind the short version — how the ghrelin receptor and growth hormone pulse actually work, what the human studies measured, what selectivity does and does not mean, common misunderstandings, who it is not for in full, and what your consultation covers.

What it is

Ipamorelin is a very short synthetic peptide — five amino acids — that acts at the ghrelin receptor in the pituitary gland, prompting a pulse of the body's own growth hormone rather than supplying growth hormone from an outside source.

It is used by people interested in body composition, recovery, and sleep quality, usually taken at night to sit alongside the body's own overnight growth hormone release. It is considered the most selective of the growth hormone secretagogues, which is part of why it is the one most often chosen.

What people use it for

USED FOR	EVIDENCE	WHAT THAT MEANS
Releasing your own growth hormone	**Human trials**	The one effect measured directly in people — it does what it is designed to do.
Lean mass and body composition	**Theoretical**	Follows from the growth hormone rise rather than from a trial that measured it.
Sleep quality and recovery	**Theoretical**	Commonly reported and consistent with the mechanism; not formally studied.

How it works

Ipamorelin binds to the ghrelin receptor (also called the growth hormone secretagogue receptor) in the pituitary gland, triggering the pituitary to release a pulse of the body's own growth hormone. This differs from directly injecting growth hormone itself — it works by stimulating the body's own release mechanism.

Compared with older growth hormone secretagogues, Ipamorelin is described as more selective for growth hormone release, with less effect on cortisol, prolactin, and appetite-related hormones — which is the basis for it being considered the most selective compound in its class.

What the research actually looked at

The growth-hormone-releasing effect of Ipamorelin has been measured directly in human studies — this is the one effect that has actually been tested in people, and the peptide reliably does what it is designed to do at the level of the pituitary pulse.

Claims about lean mass, body composition, sleep quality, and recovery follow logically from the growth hormone increase but have not themselves been measured in a published human trial. Essentially all of the human data on Ipamorelin comes from two published studies, the

principal one a 1999 trial in 40 healthy men given the compound intravenously — a different route than the injectable form used today.

Common misunderstandings

- **“It's proven to improve body composition and sleep.”** Only the growth-hormone-releasing effect has been measured directly in people; the downstream body-composition and sleep claims follow from that mechanism but have not themselves been tested in a published human trial.
- **“It's an approved medication.”** Ipamorelin has never been approved in any country and has no established quality standard; FDA has placed ipamorelin acetate on a list of bulk substances that may present significant safety risks.
- **“The human data reflects how it's actually used today.”** The principal human study used intravenous administration in healthy men in 1999 — a different route from the subcutaneous injections used in current practice.
- **“Being selective means it's risk-free.”** Selectivity for growth hormone release over other hormones does not eliminate the risks associated with elevating growth hormone itself, including effects on blood sugar.

Who it is not for

- Anyone with an active or recent cancer.
- Anyone with diabetes or prediabetes — growth hormone raises blood sugar.
- Anyone pregnant, trying to conceive, or breastfeeding.
- Children and adolescents whose growth plates are still open.

Competitive athletes: Growth hormone-releasing peptides, including Ipamorelin, fall within WADA's S2 category of prohibited peptide hormones, growth factors, and related substances; they are prohibited at all times in tested sport.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

Where this stands

Never approved, anywhere. No approval in any country, no quality standard. FDA has placed ipamorelin acetate on its list of bulk substances that may present significant safety risks, citing a published trial in which serious adverse events occurred.

Two published human studies, both intravenous. Essentially all human data comes from two studies, the main one a 1999 trial in 40 healthy men given the drug intravenously. No published human trial of the injected form used today has measured body composition, recovery or sleep.

It does reliably raise growth hormone — that part is measured. Whether raising it produces the outcomes people are after has not been studied in people.

Clinical detail

A five-amino-acid synthetic peptide acting as a selective agonist at the ghrelin (growth hormone secretagogue) receptor.

Considered the most selective growth hormone secretagogue in its class, with minimal reported effect on cortisol and prolactin relative to older secretagogues.

Principal human study: a 1999 trial in 40 healthy men, administered intravenously.

FDA has listed ipamorelin acetate among bulk substances that may present significant safety risks, citing a published trial with serious adverse events.

No published human trial of the subcutaneous form used in current practice has measured body composition, recovery, or sleep outcomes.

The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.

Ipamorelin is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.

Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose.

Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.